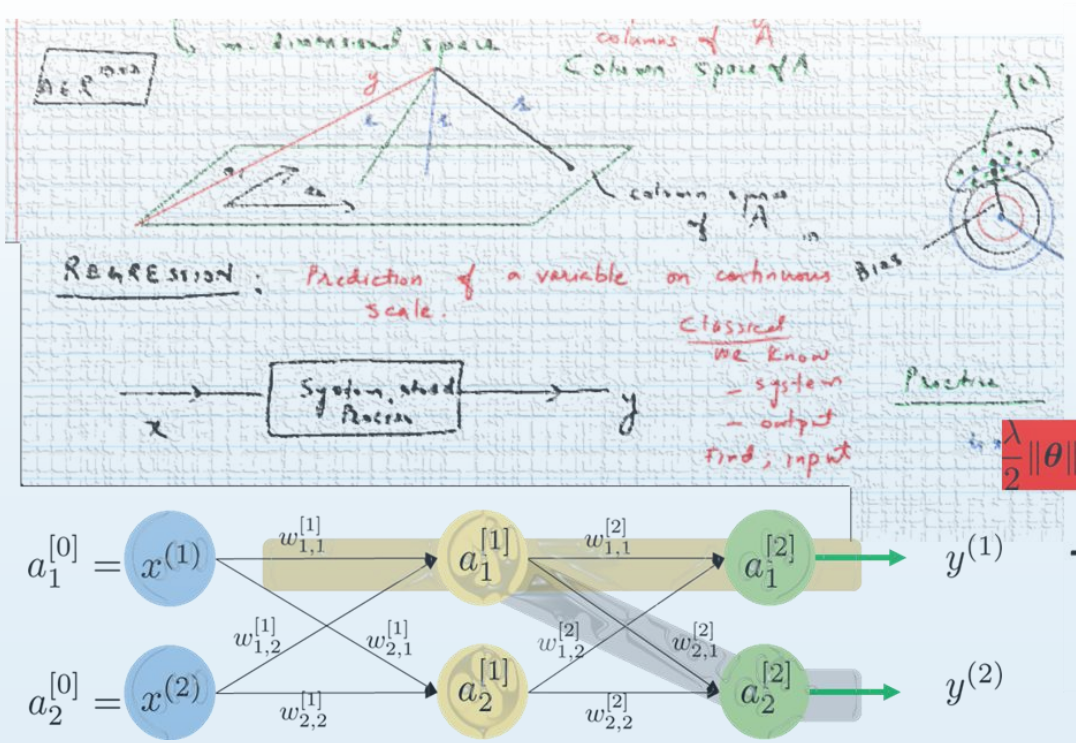
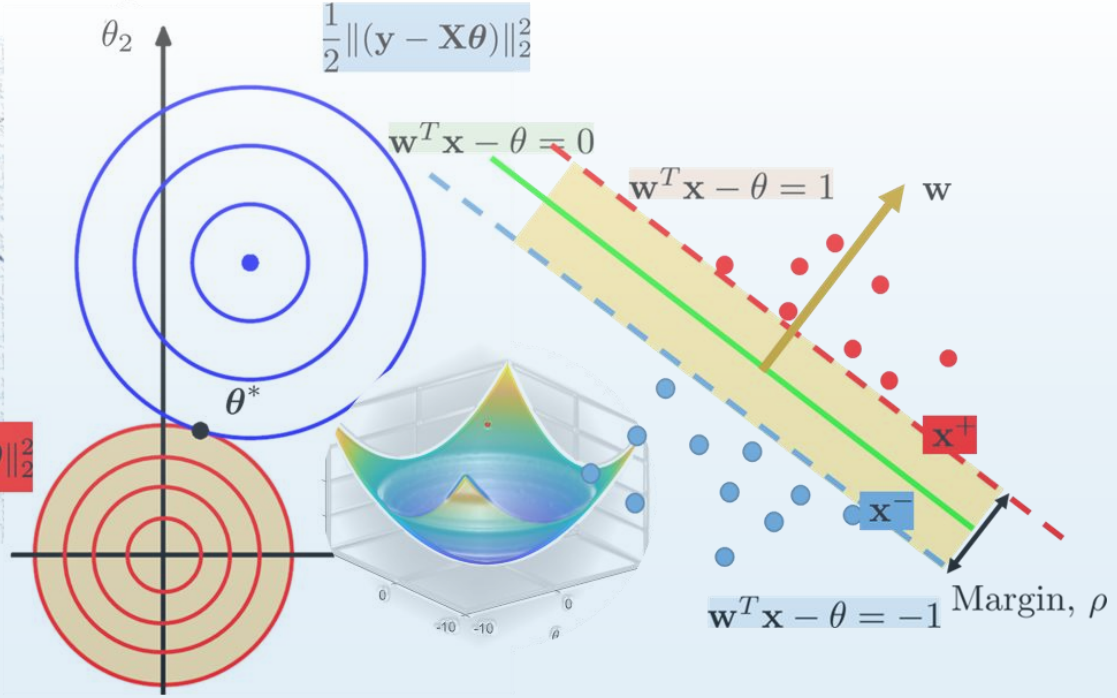




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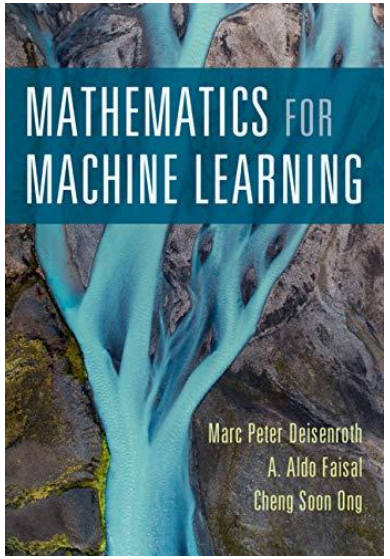
Eigenvalue Decomposition

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Outline

- *Eigenvalue Decomposition*
  - *Eigenvectors, Eigenvalue overview*
  - *Formulation*
  - *Interpretation*



Chapter 4

Eigenvalue Decomposition (EVD)

Eigenvectors and Eigenvalues:

For square matrices, *eigenvectors* and *eigenvalues* are vectors and numbers represent the *eigen-decomposition* of a matrix; analyzes the structure of this matrix.

For a matrix $A \in \mathbf{R}^{n \times n}$

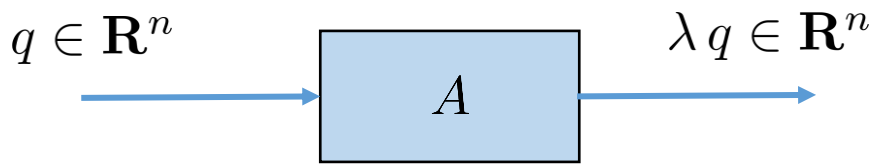
a vector $q \in \mathbf{R}^n$, $q \neq 0$, is called an eigenvector of A if

$$A\,q = \lambda\,q$$ Eigenvalue Equation

- λ is referred to as an eigenvalue of A associated with the eigenvector q

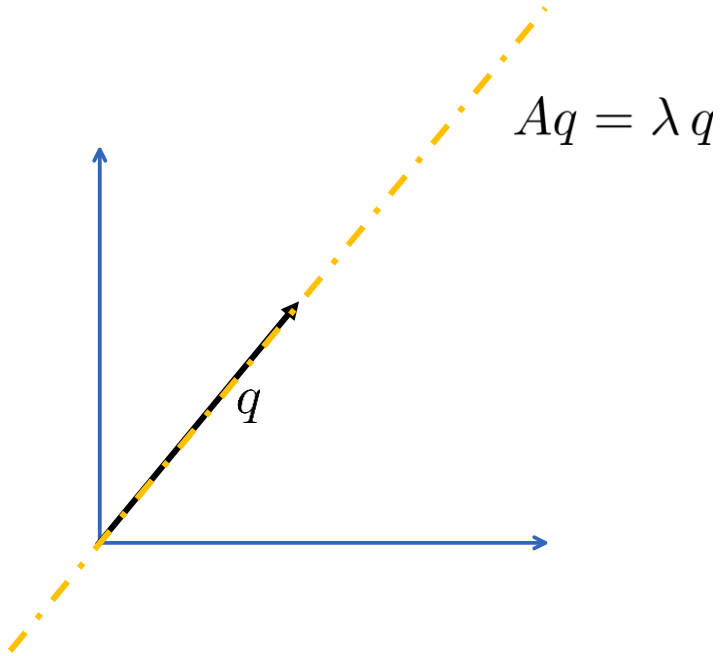
Eigenvalue Decomposition (EVD)

Linear transformation interpretation:



Only scaling

Graphically



Eigenvalue Decomposition (EVD)

Eigenvectors and Eigenvalues:

How to compute eigenvalues?

$$A\,q = \lambda\,q \qquad \Rightarrow \quad Aq - \lambda q = 0$$

$p(\lambda) = \det(Aq - \lambda q) = 0$ Characteristic polynomial; degree *n*

n eigenvalues Eigenspectrum: set of eigenvalues

n eigenvectors Eigenspace: span of eigenevectors

Eigenvalue Decomposition (EVD)

Eigenvectors and Eigenvalues:

$$A\,q = \lambda\,q$$

n eigenvalues $\quad \lambda_1,\, \lambda_2, \ldots, \lambda_n$

n eigenvectors $\quad q_1,\, q_2, \ldots, q_n \in \mathbf{R}^n$

$$A\,q_1 = \lambda_1\,q_1, \quad A\,q_2 = \lambda_2\,q_2, \ \ldots$$

$$A\,Q = Q\,\Lambda$$

$$A = Q\,\Lambda\,Q^{-1}$$

Eigen-decomposition

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \lambda_n \end{bmatrix}$$

$$Q = \begin{bmatrix} (q_1)_1 & (q_2)_1 & \ldots & (q_n)_1 \\ (q_1)_2 & (q_2)_2 & \ldots & (q_n)_2 \\ \vdots & \vdots & \ddots & \vdots \\ (q_1)_n & (q_2)_n & \ldots & (q_n)_n \end{bmatrix}$$

Eigenvalue Decomposition (EVD)

Eigenvectors and Eigenvalues – Numerical Example (Tutorial Problem):

$$A = \begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix}$$

The characteristic polynomial of A is obtained by calculating the determinant of $A - \lambda I$:

$$\det(A - \lambda I) = 0$$

Compute $A - \lambda I$:

$$A - \lambda I = \begin{bmatrix} 4-\lambda & -2 \\ -2 & 1-\lambda \end{bmatrix}$$

Compute the determinant:

$$\begin{aligned} \det(A - \lambda I) &= (4-\lambda)(1-\lambda) - (-2)(-2) \\ &= (4-\lambda)(1-\lambda) - 4 \\ &= (4 \times 1 - 4\lambda - \lambda \times 1 + \lambda^2) - 4 \\ &= (4 - 4\lambda - \lambda + \lambda^2) - 4 \\ &= (\lambda^2 - 5\lambda + 4) - 4 \\ &= \lambda^2 - 5\lambda \end{aligned}$$

Set the characteristic equation:

$$\lambda^2 - 5\lambda = 0 \qquad \lambda_1 = 0, \quad \lambda_2 = 5$$

Eigenvalue Decomposition (EVD)

Eigenvectors and Eigenvalues – Numerical Example (Tutorial Problem):

For $\lambda = 0$:
Solve $(A - 0I)\mathbf{v} = \mathbf{0}$:

$$\begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Set up the system of equations:

$$\begin{cases} 4v_1 - 2v_2 = 0 \\ -2v_1 + v_2 = 0 \end{cases}$$

From the second equation:

$$v_2 = 2v_1$$

Substitute back into the first equation to verify consistency:

$$4v_1 - 2(2v_1) = 0 \implies 4v_1 - 4v_1 = 0$$

Eigenvalue Decomposition (EVD)

Eigenvectors and Eigenvalues – Numerical Example (Tutorial Problem):

Thus, the eigenvector corresponding to $\lambda = 0$ is:

$$\mathbf{v}_1 = v_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Normalize $\mathbf{v}_1$:

$$\|\mathbf{v}_1\| = \sqrt{v_1^2 + (2v_1)^2} = v_1\sqrt{1+4} = v_1\sqrt{5}$$

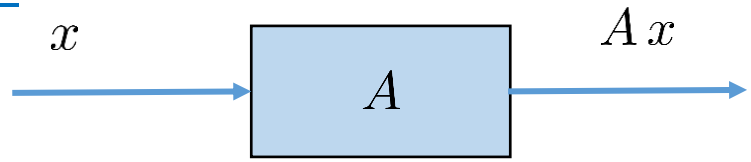
Choose $v_1 = \dfrac{1}{\sqrt{5}}$, so the normalized eigenvector is:

$$\mathbf{v}_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

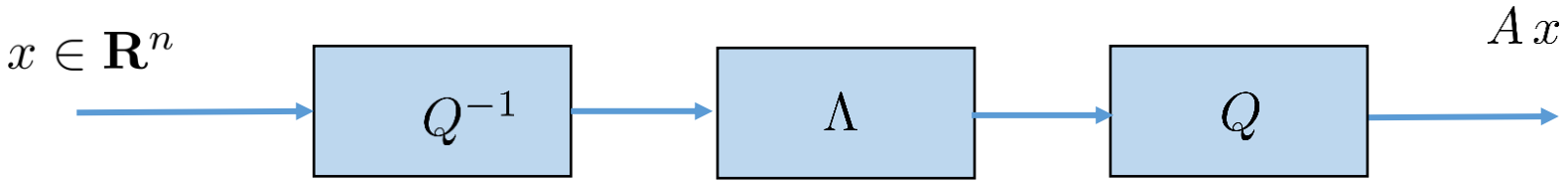
Eigenvalue Decomposition

What does eigenvector and eigenvalues reveal about A?

Linear Transformation



Linear Transformation Interpretation in terms of Eigen-decomposition of the Matrix



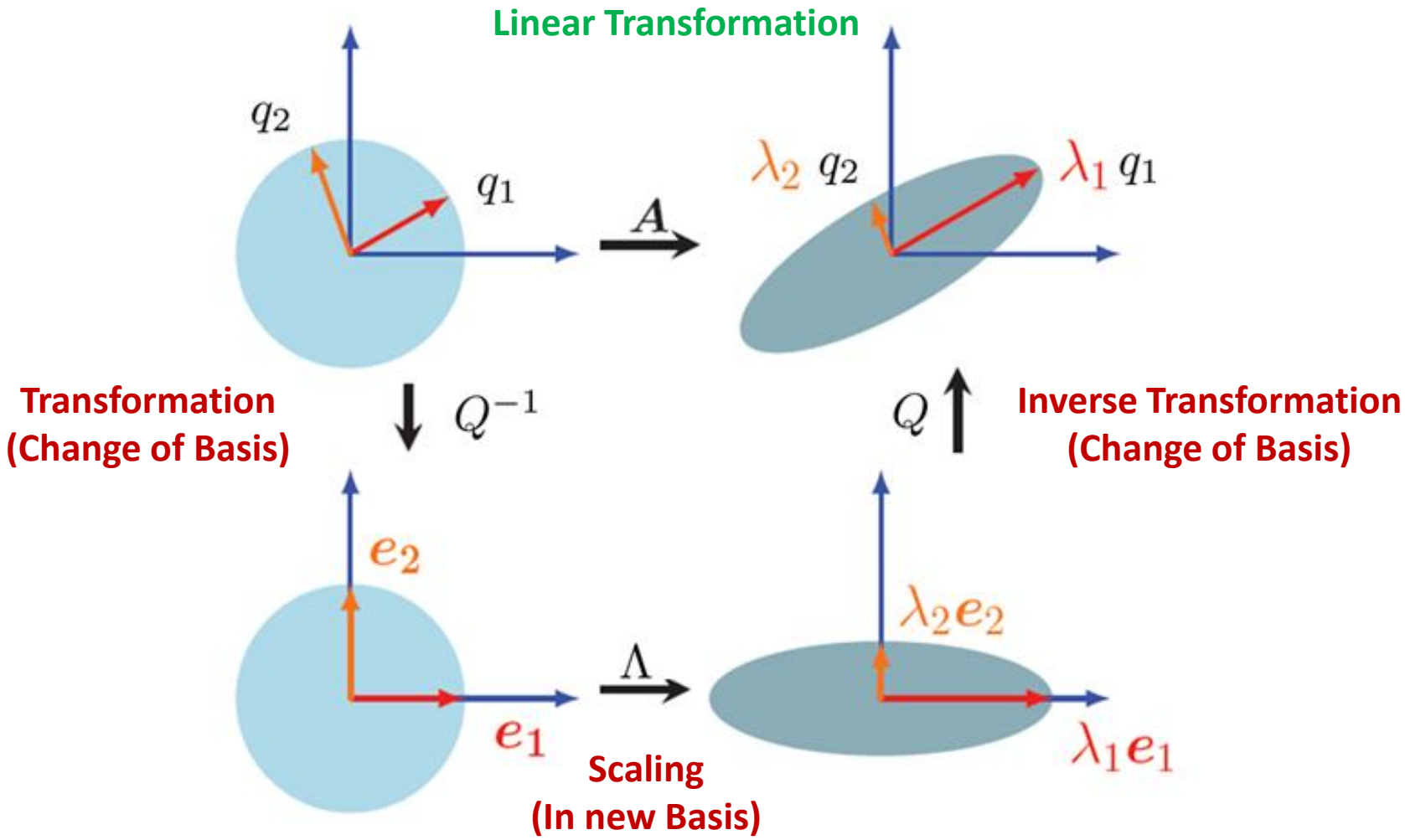
Transformation (Change of Basis)

Scaling (In new Basis)

Inverse Transformation (Change of Basis)

Eigenvalue Decomposition

Linear Transformation Interpretation in terms of Eigen-decomposition of the Matrix - Visualization



Eigenvalue Decomposition

Determinant in terms of Eigenvalues

- Determinant of a matrix $A \in \mathbf{R}^{n \times n}$ is given by the product of eigenvalues, that is,

$$\det(A) = \det(Q\Lambda Q^{-1}) = \det(Q)\det(\Lambda)\det(Q^{-1})$$

$$\det(A) = \det(Q\Lambda Q^{-1}) = \det(Q)\left(\prod_{i=1}^{n}\lambda_i\right)\frac{1}{\det(Q)}$$

$$\det(A) = \prod_{i=1}^{n}\lambda_i$$

- If one of the eigenvalues is zero, determinant of the matrix is zero.
  - I encourage you to connect this with the interpretation of determinant and (eigenvectors, eigenvalues).

Eigenvalue Decomposition (EVD)

EVD of Inverse Matrix:

- Matrix inverse A^{-1} has same eigenvectors but eigenvalues are given by inverse of the eigenvalues of the original matrix A.
- This can be shown in multiple ways. Let's use eigenvalue equation to show this.

For a matrix $A \in \mathbf{R}^{n \times n}$

$$A\,q = \lambda\,q$$ **Eigenvalue Equation**

- Assuming A is invertible, that is, all eigenvalues are non-zero.

$$A^{-1}A\,q = \lambda\,A^{-1}q \qquad \frac{1}{\lambda}q = A^{-1}q$$

This implies q is an eigenvector of A^{-1} with associated eigenvalue $\frac{1}{\lambda}$.

Eigenvalue Decomposition

Power of a matrix

$$AA = A^2 = Q\Lambda Q^{-1}\ Q\Lambda Q^{-1} = Q\Lambda^2 Q^{-1}$$

$$A^n = Q\Lambda^n Q^{-1}$$

- I encourage you to connect this with the interpretation of linear transformation using EVD of a matrix.

Eigenvalue Decomposition

Zero eigenvalues; Columns of A are not linearly independent

- If one of the eigenvalues is zero and q is an associated eigenvector.

$$A\,q = \lambda\,q \;=\; 0$$

- It simply follows from the definition of linear independence that the columns are not linearly independent since Aq represents the linear combination of columns of A.

Eigenvalue Decomposition

Eigenvalues of a Symmetric Matrix

(Spectral Theorem) For a symmetric matrix $A \in \mathbf{R}^{n \times n}$, there exists an orthonormal basis of the corresponding vectors space consisting of eigenvectors of A and each eigenvalue is real.

$$A = Q \Lambda Q^{-1}$$

$$A^T = \left(Q^{-1}\right)^T \Lambda Q^T$$

Since $A = A^T$, we have $Q^{-1} = Q^T$

⇓

Orthonormal matrix

- Summary: For a symmetric matrix, eigenvectors are orthonromal and eigenvalues are real.